

class09

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2 Importing candy data

```
#load dataset
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-rankings.csv"

#inspect data
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

A1. There are 85 different candy types in this dataset.

```
dim(candy)
```

```
[1] 85 12
```

Q2. How many fruity candy types are in the dataset?

A2. There are 38 fruity candy types in the dataset.

```
sum(candy$fruity)
```

```
[1] 38
```

```
# find the winpercent value for Twix  
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
#alternate way to find the winpercent value using dplyr  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>  
  filter(row.names(candy)=="Twix") |>  
  select(winpercent)
```

```
      winpercent  
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

A3. My favorite candy is Almond Joy. Its winpercent value is 50.34755.

```
# find the winpercent value of Almond Joy
candy["Almond Joy",]$winpercent
```

```
[1] 50.34755
```

Q4. What is the winpercent value for “Kit Kat”?

A4. The winpercent value is 76.7686.

```
# find the winpercent value of Kit Kat
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

A5. The winpercent value is 49.6535.

```
# find the winpercent value of Tootsie Rolls
candy["Tootsie Roll Snack Bar",]$winpercent
```

```
[1] 49.6535
```

```
# load skimr package
library("skimr")
# look at skimr dataset overview
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

A6. Yes, winpercent is the only variable that contains percentages (on a scale from 0 to 100) rather than decimals (on a scale from 0 to 1).

Q7. What do you think a zero and one represent for the candy\$chocolate column?

A7. A zero represents that the candy does not contain chocolate, while a one represents that the candy does contain chocolate.

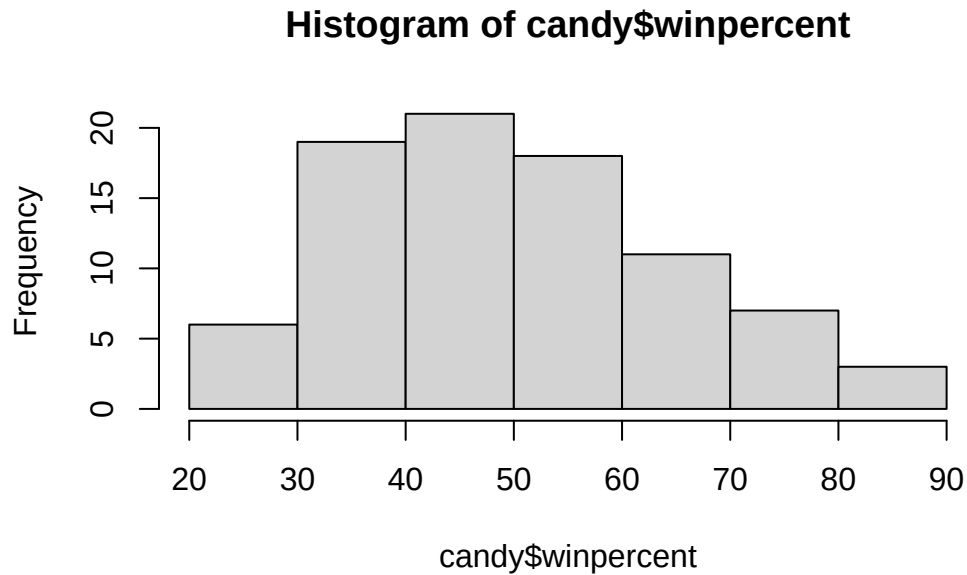
```
candy$chocolate
```

```
[1] 1 1 0 0 0 1 1 0 0 0 1 0 0 0 0 0 0 0 0 0 1 1 1 1 0 1 1 0 0 0 1 1 0 1 1 1
[39] 1 1 1 0 1 1 0 0 0 1 0 0 0 1 1 1 1 0 1 0 0 1 0 0 1 0 1 1 0 0 0 0 0 0 0 1 1
[77] 1 1 0 1 0 0 0 0 1
```

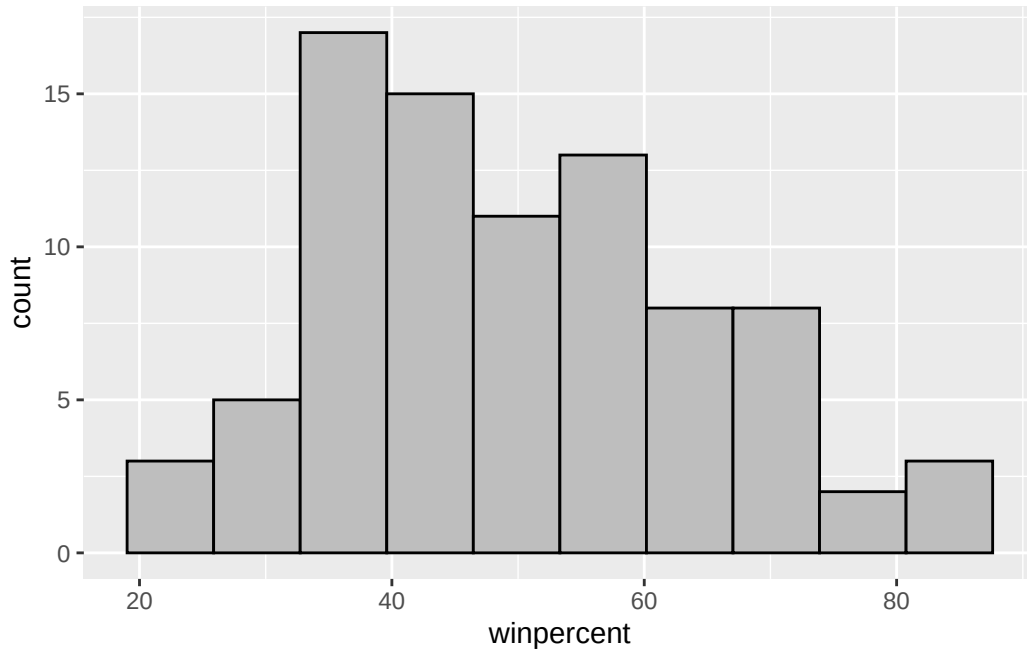
3 Exploratory analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
# create histogram of winpercent values using base R  
hist(candy$winpercent)
```



```
# load ggplot2  
library(ggplot2)  
  
# create histogram of winpercent values using ggplot  
ggplot(candy) +  
  aes(winpercent) +  
  geom_histogram(bins = 10, fill="grey", color = "black")
```



Q9. Is the distribution of winpercent values symmetrical?

A9. No, the distribution is not symmetrical. It seems to be skewed right, with the maximum frequency winpercent values being at the 40-50 bar.

Q10. Is the center of the distribution above or below 50%?

A10. No, the center of distribution (median) is 47.83%, thereby indicating it is below 50%.

```
# observe distribution of data
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

A11. On average, chocolate candy is higher ranked than fruity candy, with a mean winpercent value of 60.92%. Fruity candy has a mean value of 44.12%.

```
# t-test of chocolate
chocolate <- candy$winpercent[as.logical(candy$chocolate)]
t.test(chocolate)
```

One Sample t-test

```
data: chocolate
t = 28.926, df = 36, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 56.65009 65.19297
sample estimates:
mean of x
 60.92153
```

```
#t-test of fruity candy
fruit <- candy$winpercent[as.logical(candy$fruity)]
t.test(fruit)
```

One Sample t-test

```
data: fruit
t = 26.498, df = 37, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 40.74612 47.49337
sample estimates:
mean of x
 44.11974
```

Q12. Is this difference statistically significant?

A12. Yes, this difference is statistically significant, as the data has a $p\text{-value} < 0.05$. Therefore, it is extremely unlikely that the results are based on random chance.

4. Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

A13. The five least liked candy types are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```
# look at the top five lowest ranked candy types
head(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

A14. The five most liked candy types are Snickers, Kit Kat, Twix, Reese's Miniatures, and Reese's Peanut Butter cup.

```
# look at the top five highest ranked candy types
tail(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Snickers	1	0	1		1	1
Kit Kat	1	0	0		0	0
Twix	1	0	1		0	0
Reese's Miniatures	1	0	0		1	0
Reese's Peanut Butter cup	1	0	0		1	0

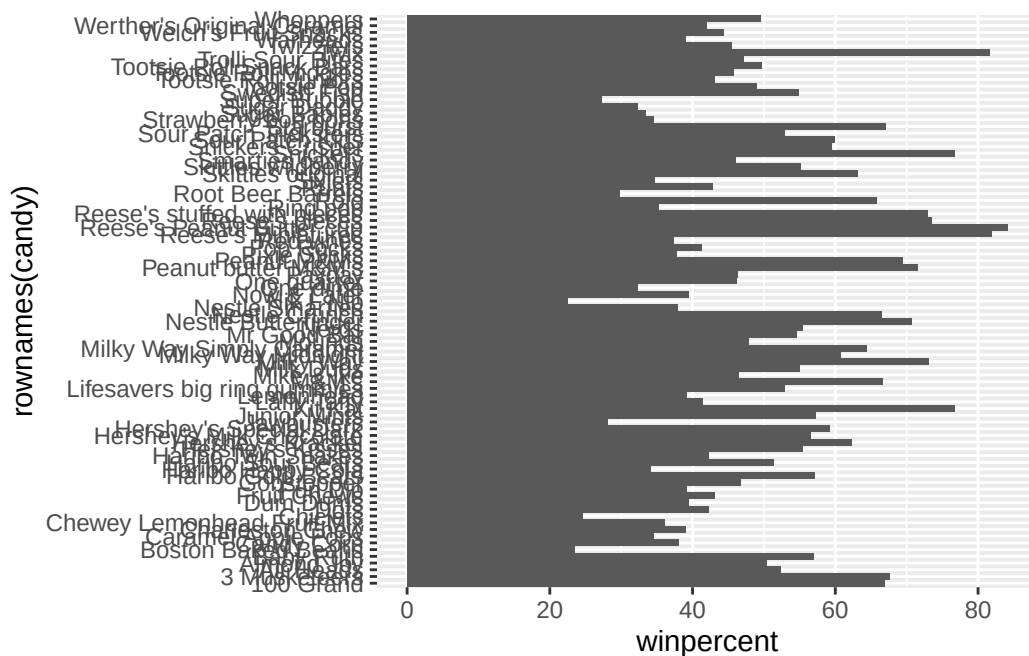
	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Snickers				0	0	1	0	0.546
Kit Kat				1	0	1	0	0.313

Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

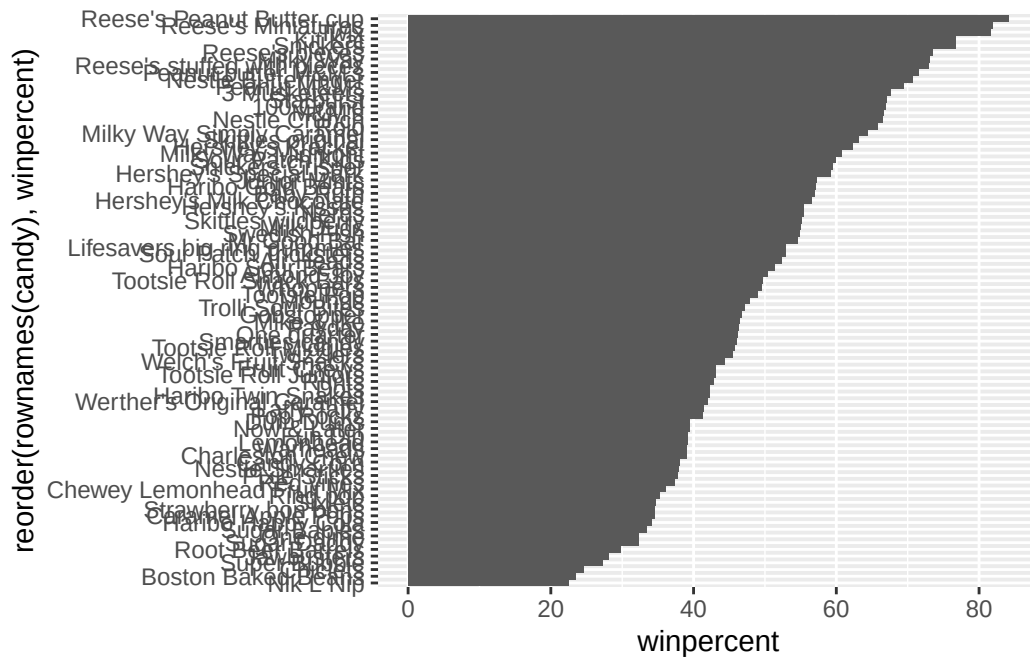
Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)
#create barplot using winpercent values
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



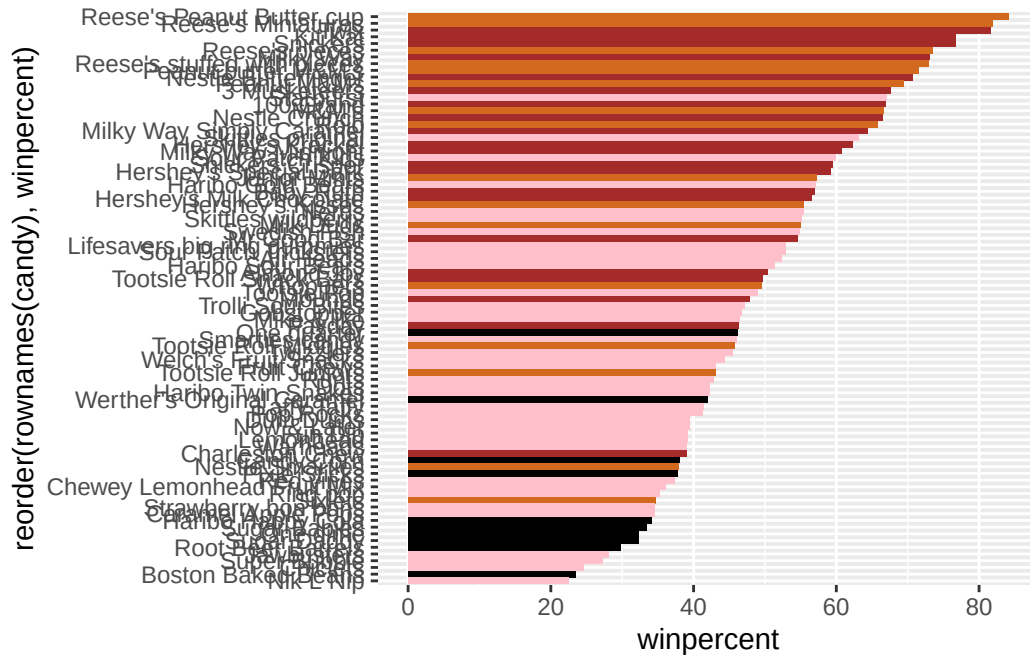
Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
#reorder based on winpercent (highest to lowest)
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col()
```



```
# add specific color for each type of candy
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
# create barplot with our color-coded candies
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

A17. The worst ranked chocolate candy is Sixlets.

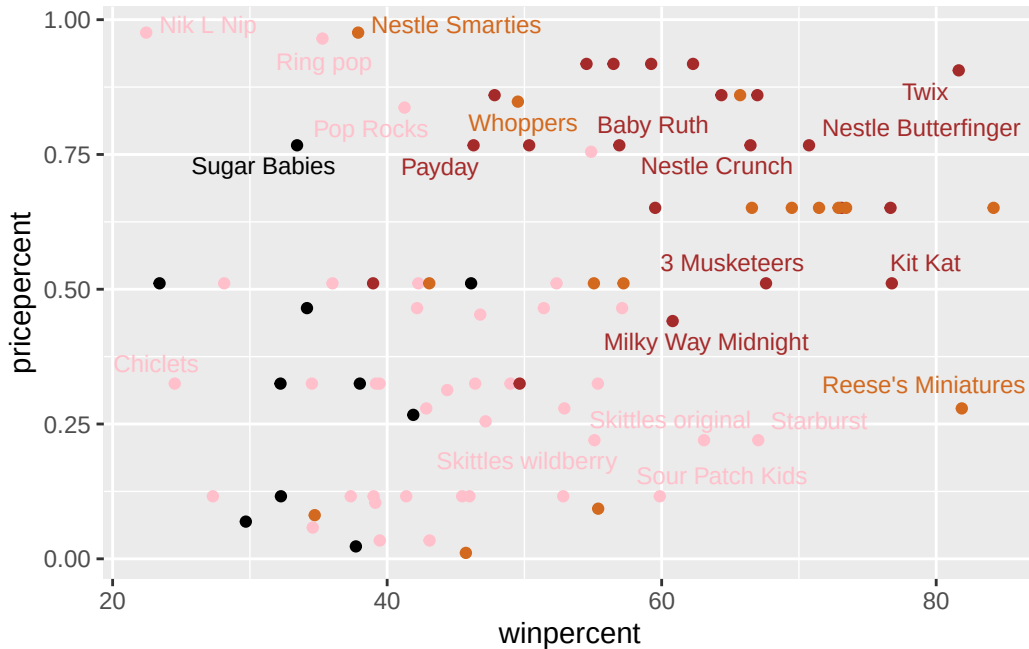
Q18. What is the best ranked fruity candy?

A18. The best ranked fruity candy is Starburst.

5. Taking a look at pricepercent

```
# load ggrepel
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

A19. The highest ranked and cheapest is Reese's Miniatures.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

A20. The five most expensive candy types are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these, the least popular is Nik L Nip.

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

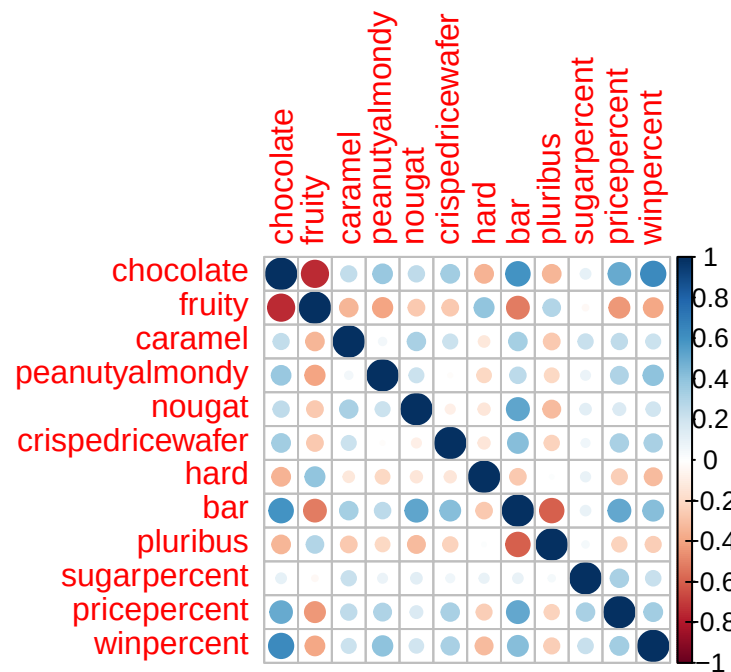
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

6. Exploring the correlation structure

```
# load corrplot  
library(corrplot)
```

corrplot 0.95 loaded

```
# create a correlation matrix of the candy dataset  
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

A22. Chocolate and fruity are the most anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

A23. Chocolate and bars are the most positively correlated.

7. Principal Component Analysis

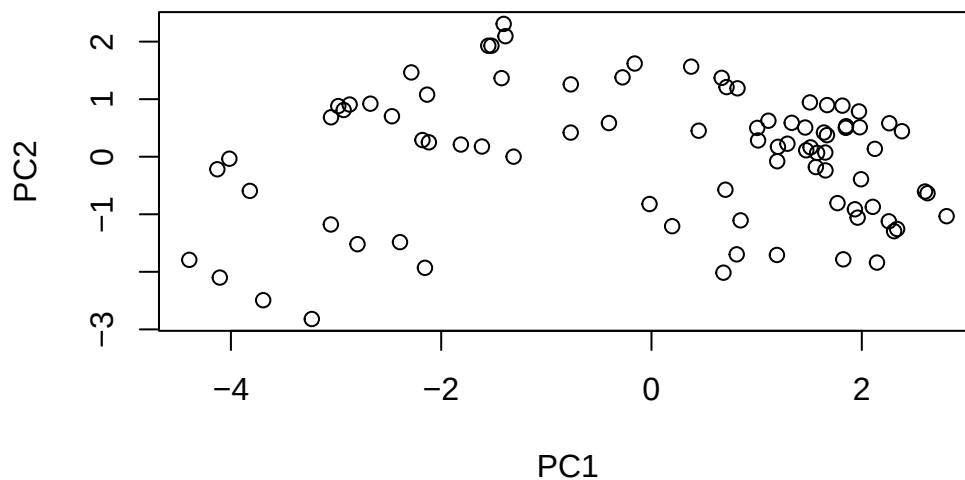
```
# apply PCA to our candy dataset
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

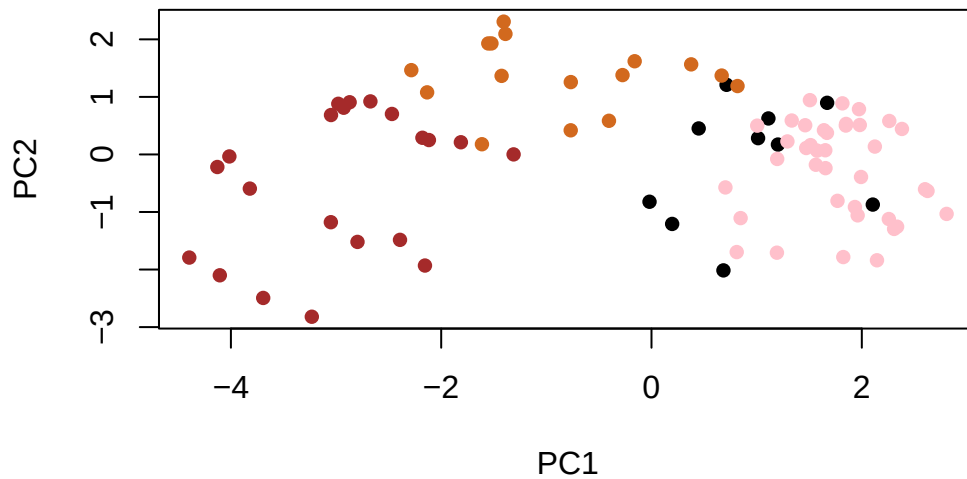
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
# plot main PCA score plot of PC1 vs. PC2
plot(pca$x[, 1:2])
```



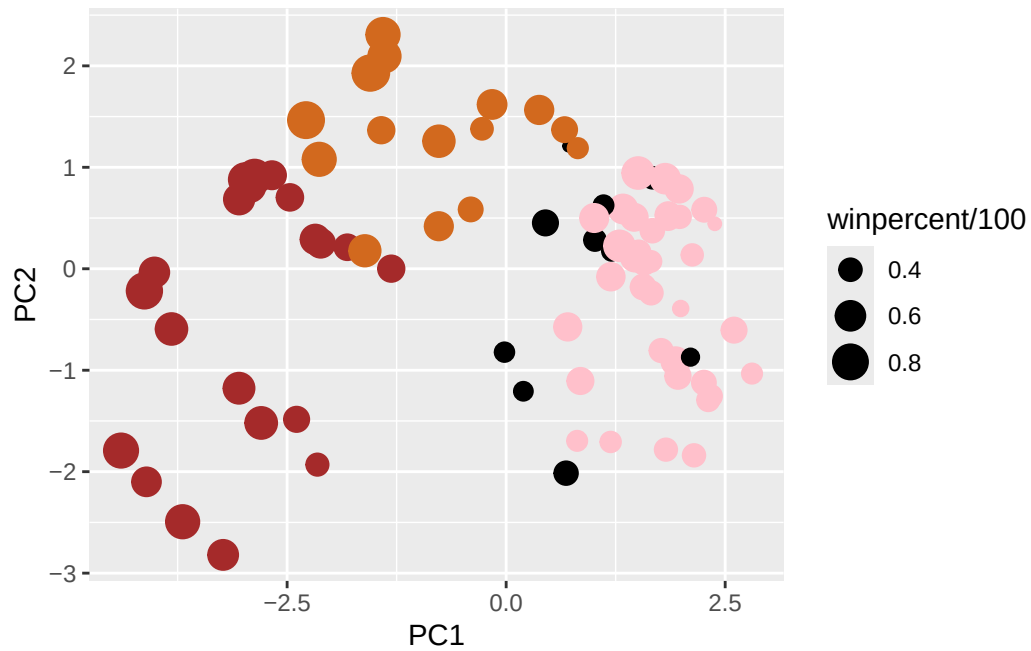
```
# change plotting character and add color
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
# change plot variables
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

p

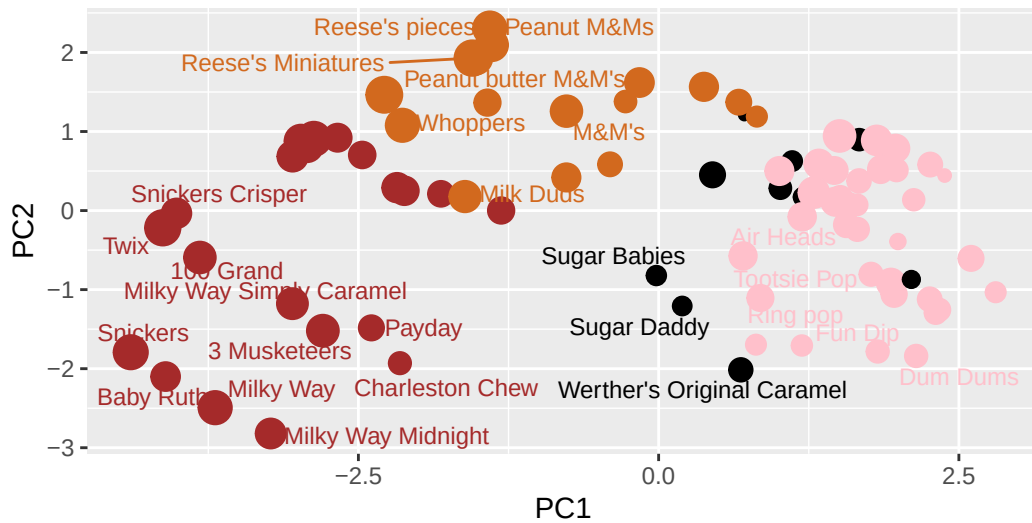


```
# load ggrepel
library(ggrepel)

# label the plot, add title and subtitle
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```


Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
# generate interactive plot to see labels  
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

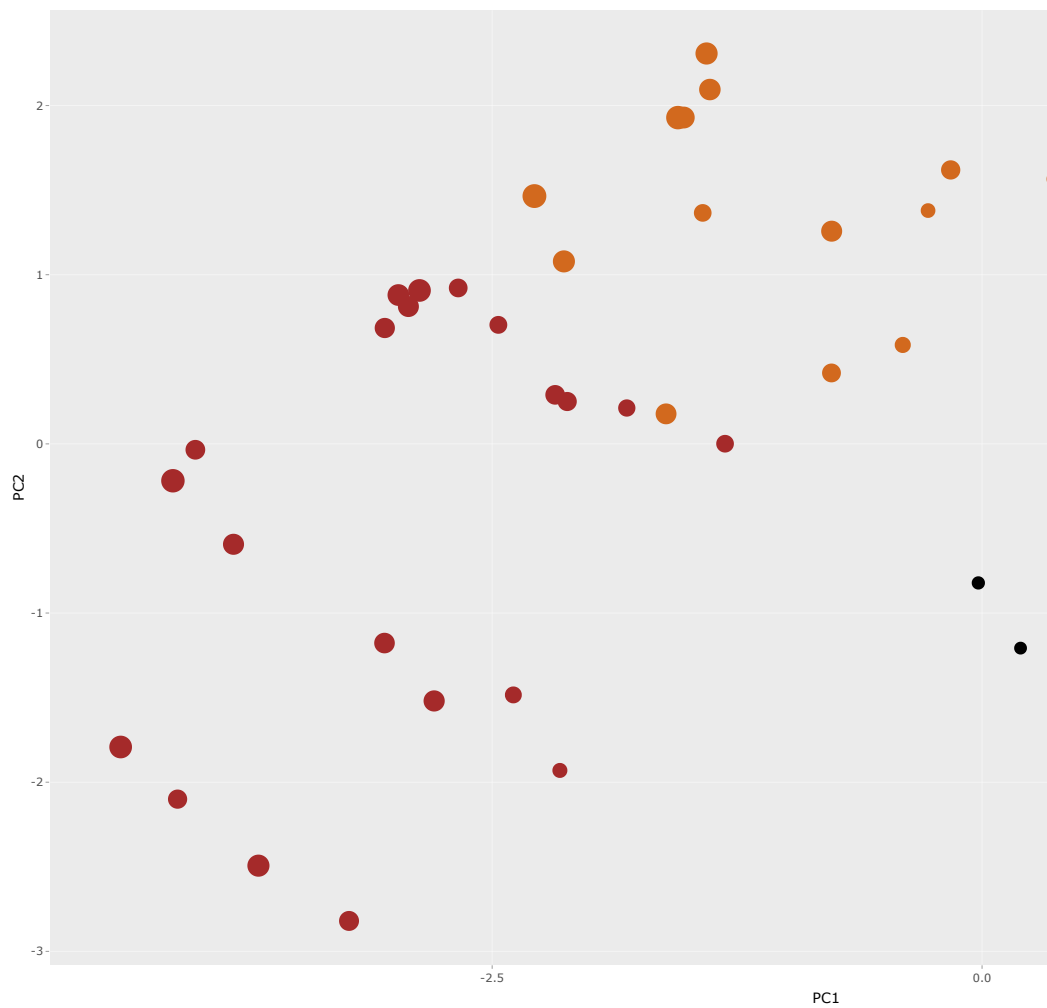
The following object is masked from 'package:stats':

filter

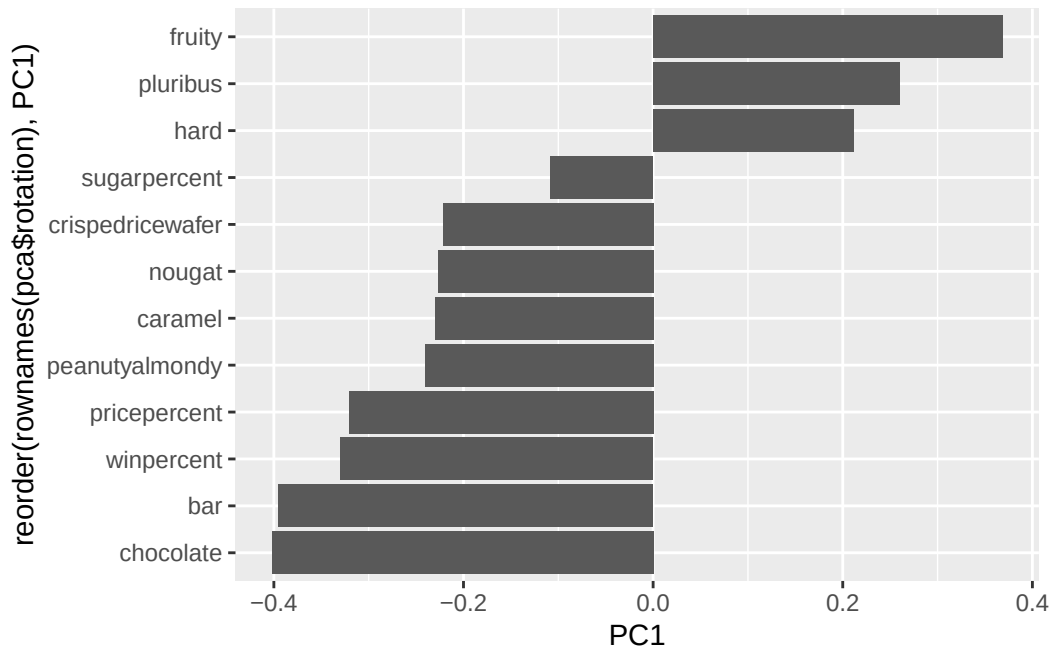
The following object is masked from 'package:graphics':

layout

```
ggplotly(p)
```



```
# create PCA loadings plot
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

A24. The variables picked up strongly in the positive direction are fruity, pluribus, and hard. The variables picked up strongly in the negative direction are chocolate and bar. These two types of variables have a very high negative correlation when put together, as we can see in the corrplot above. When determining the relationships between chocolate and the three positive variables, they are all orange/red. This means that PC1 is showing the extreme differences in the beginning/end of the plot, fruity and chocolate, and then gradually shows more positive correlations, and subsequently other types of candy, towards the middle.

8. Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A25. Chocolate + bar make a “winning” candy, as does fruity + pluribus + hard. Both chocolate and bar show up together at the end of the PCA barplot, while fruity, pluribus, and hard show up together at the beginning of the barplot, therefore indicating that these characteristics are often grouped together. The corrplot can also attribute to this, as the chocolate + bar combination have the the highest positive correlation, while the fruity + hard combination also also pretty high. Both types of the analyses show similar ideas, with the corrplot also visualizing the intensity of the correlations, and the PCA barplot focusing on the variance.